

Functional requirements:

user should be able to search for problem based on his preference,
user should be able to solve & submit solution based on his preference,
user should be able to participate in a contest based on his preference.

Non-functional requirements:

Data-partitioning,
load balancing,
fault tolerance,
vertical scalability,
horizontal scalability,
low latency ,
high throughput ,

CAP theorem (Here we need to prioritise consistency for rank predictions during coding contests & for search scenario we need to prioritise 99.99% uptime i.e Availability >> Consistency)

Core Entities :

user ,
problem,
contest,
solution

API :

POST:/submit

```
{  
  userID,  
  filemetadata  
}
```

GET:/search/{problemID}

HLD :

